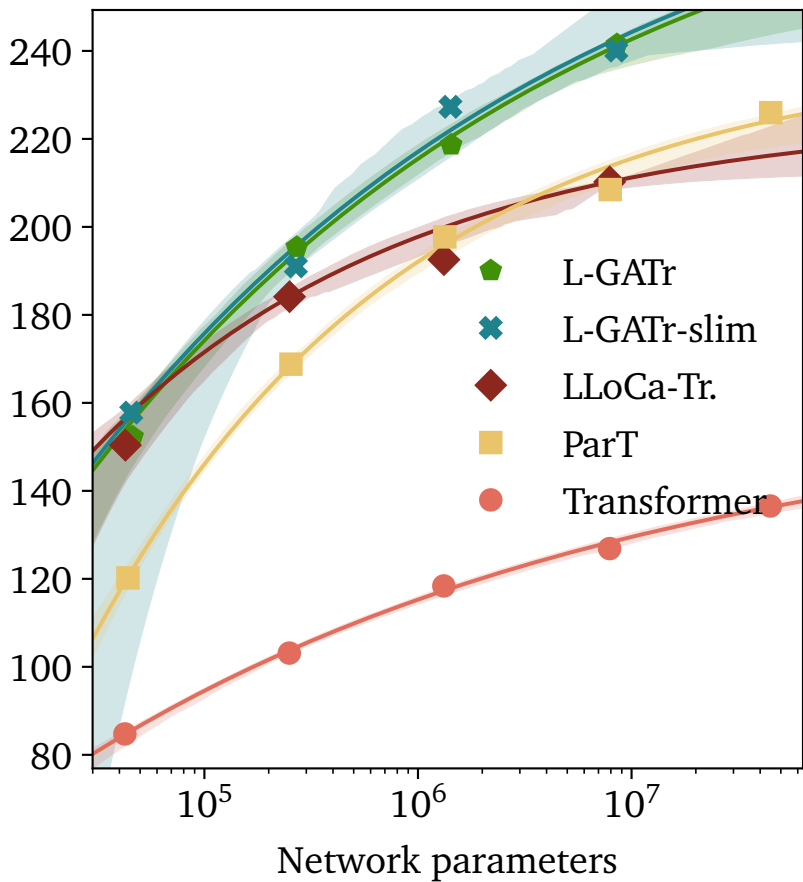
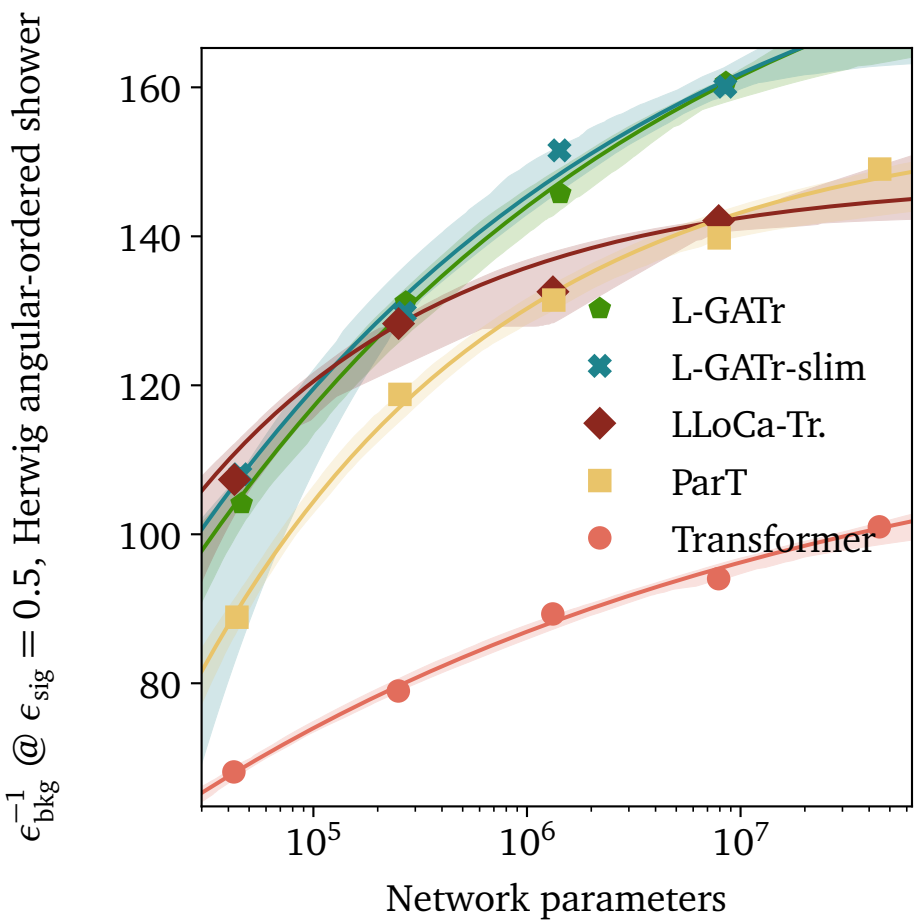
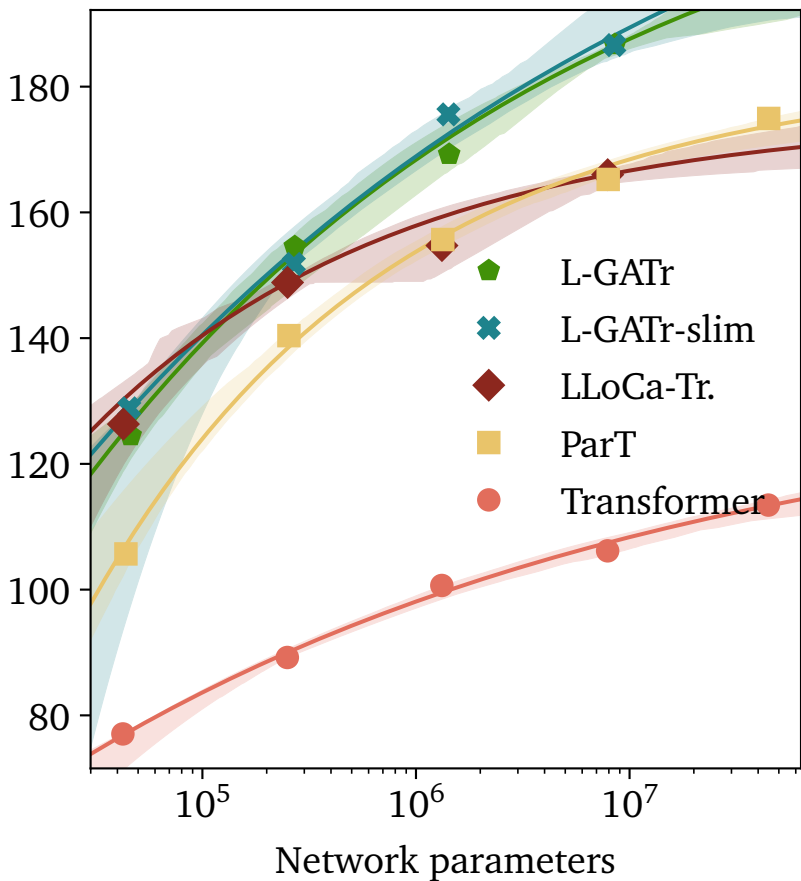


$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ nominal}$

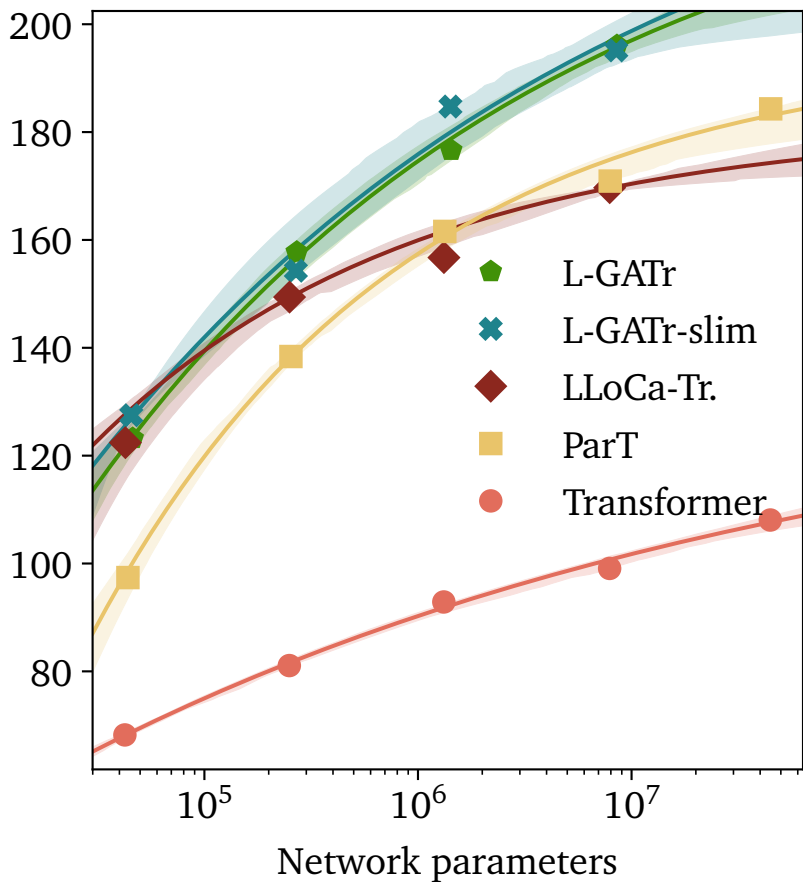




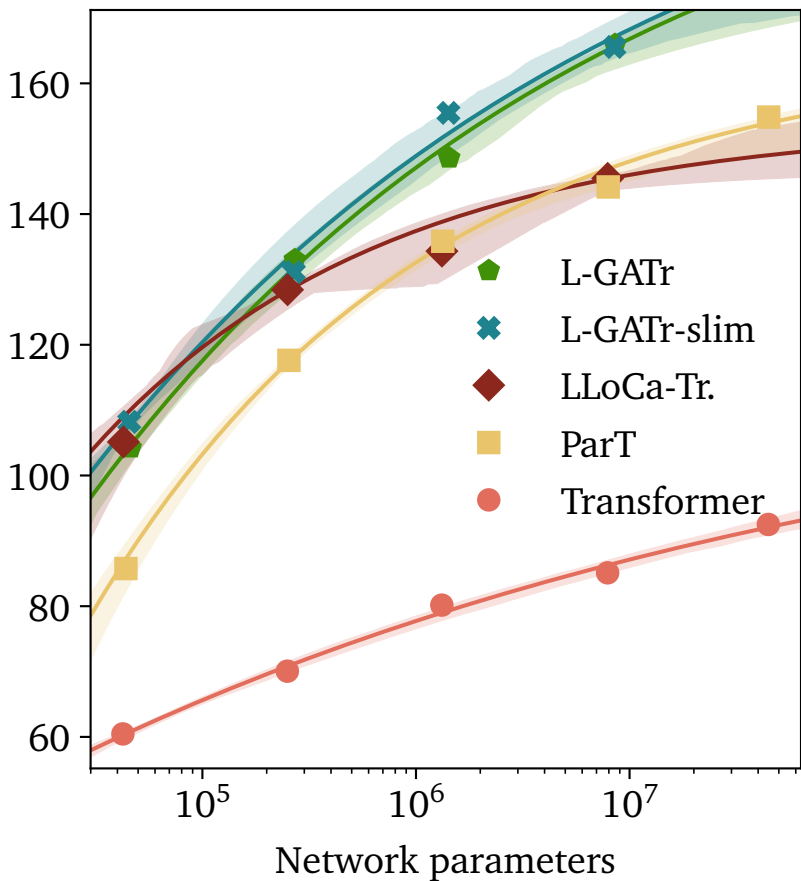
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ Herwig dipole shower}$



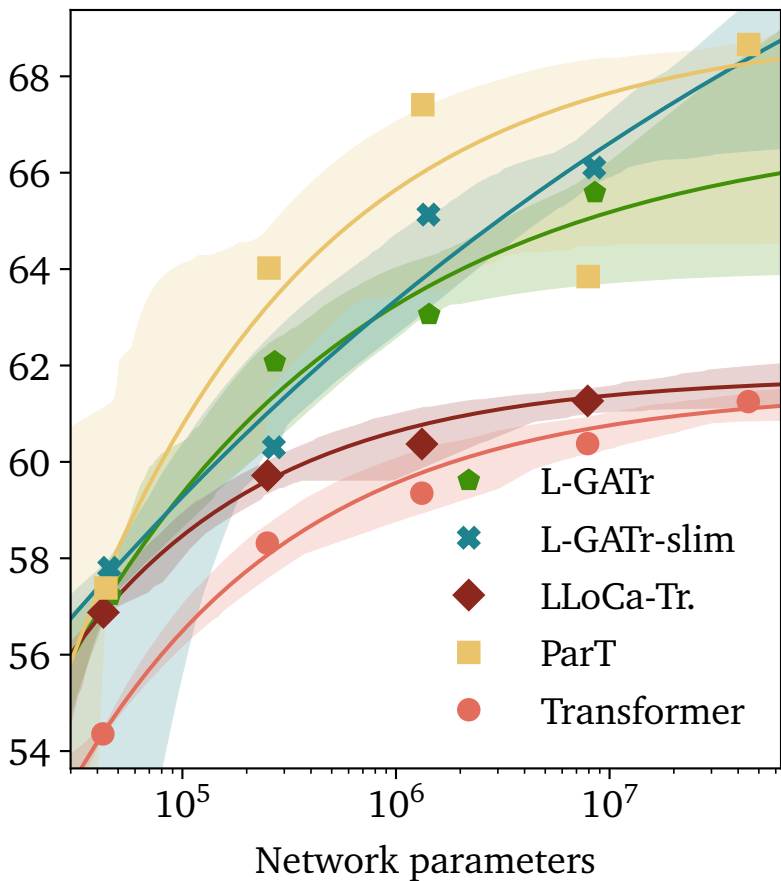
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5$, Lund string hadronization



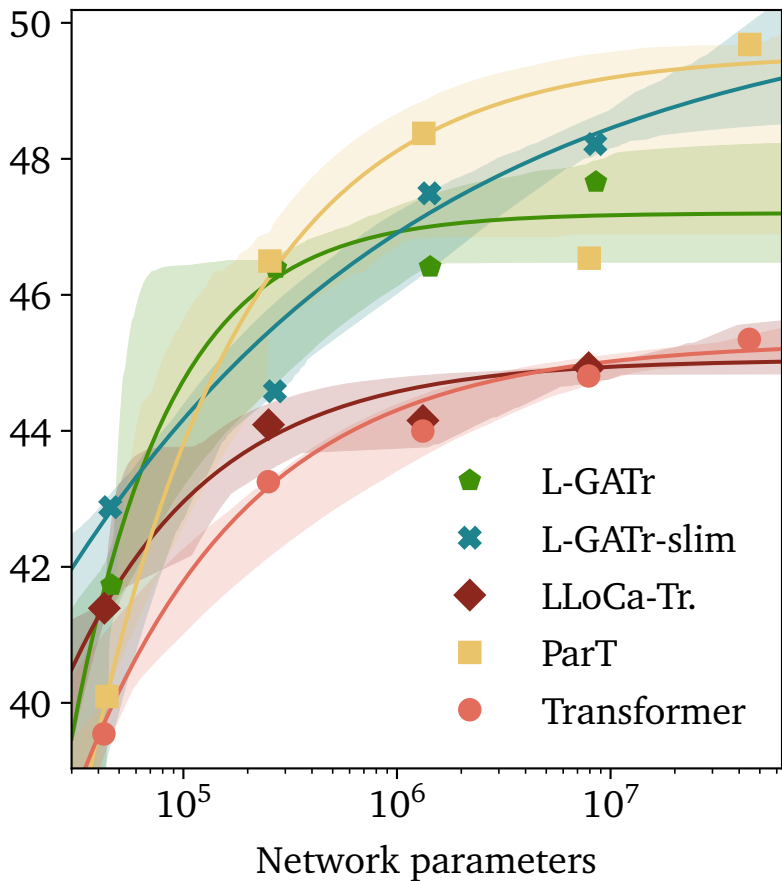
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5$, Sherpa cluster hadronization



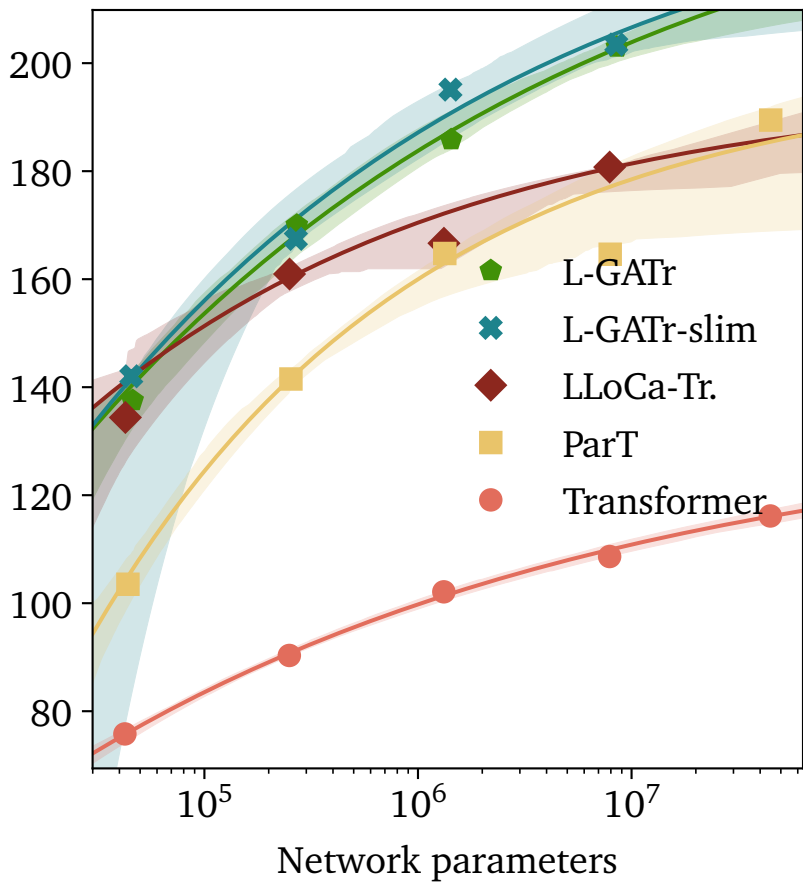
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{Pythia } t\bar{t} \text{ signal}$



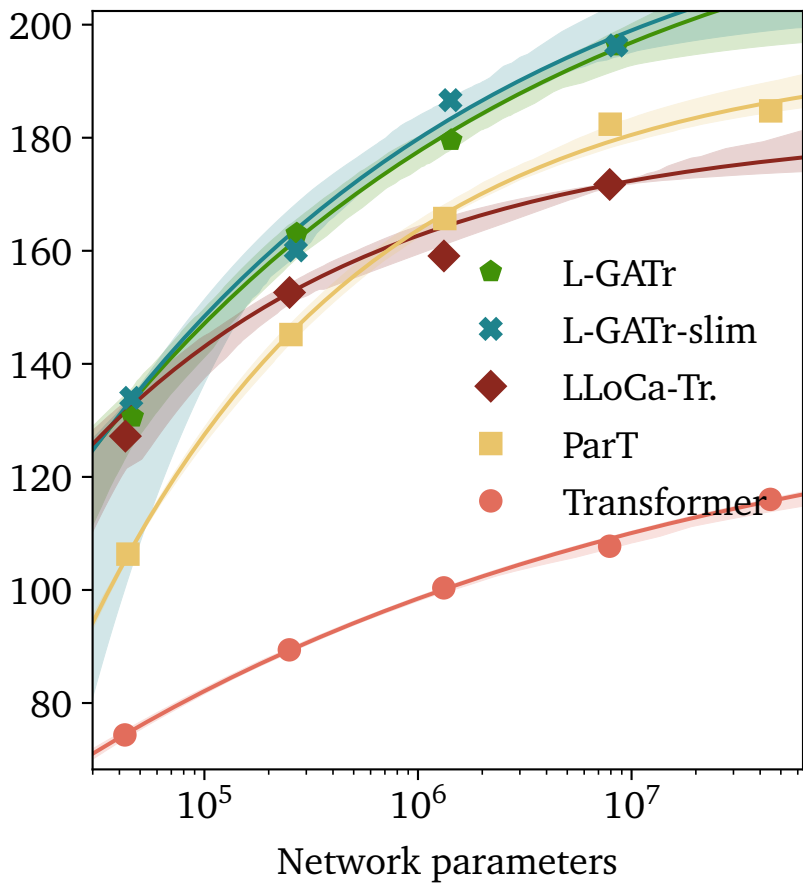
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ Herwig } t \bar{t} \text{ signal}$



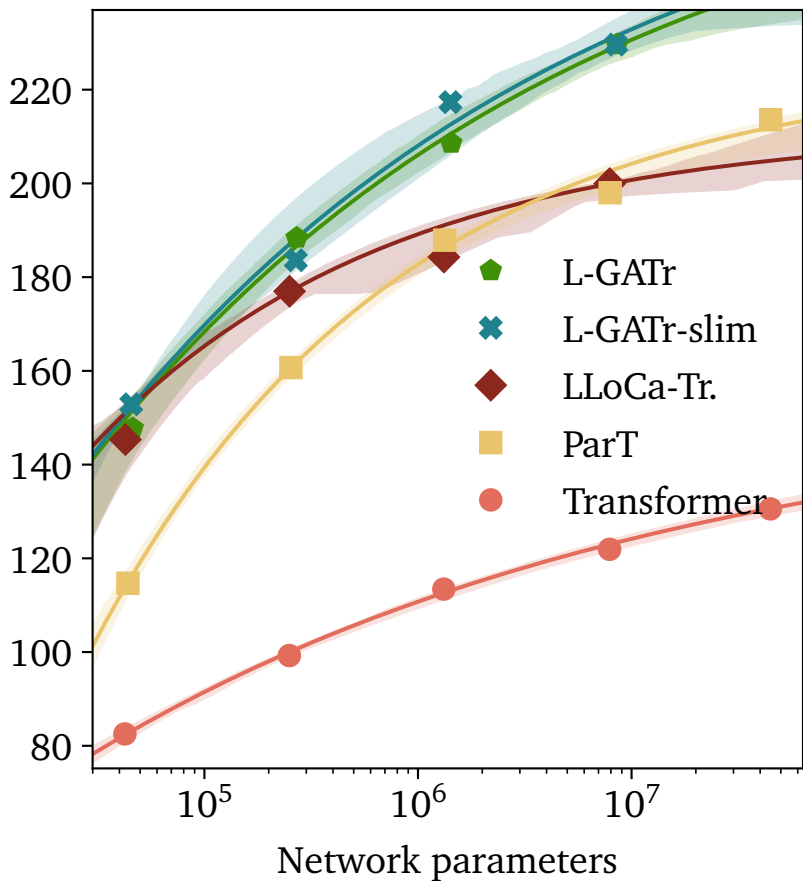
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5$, cluster energy scale up



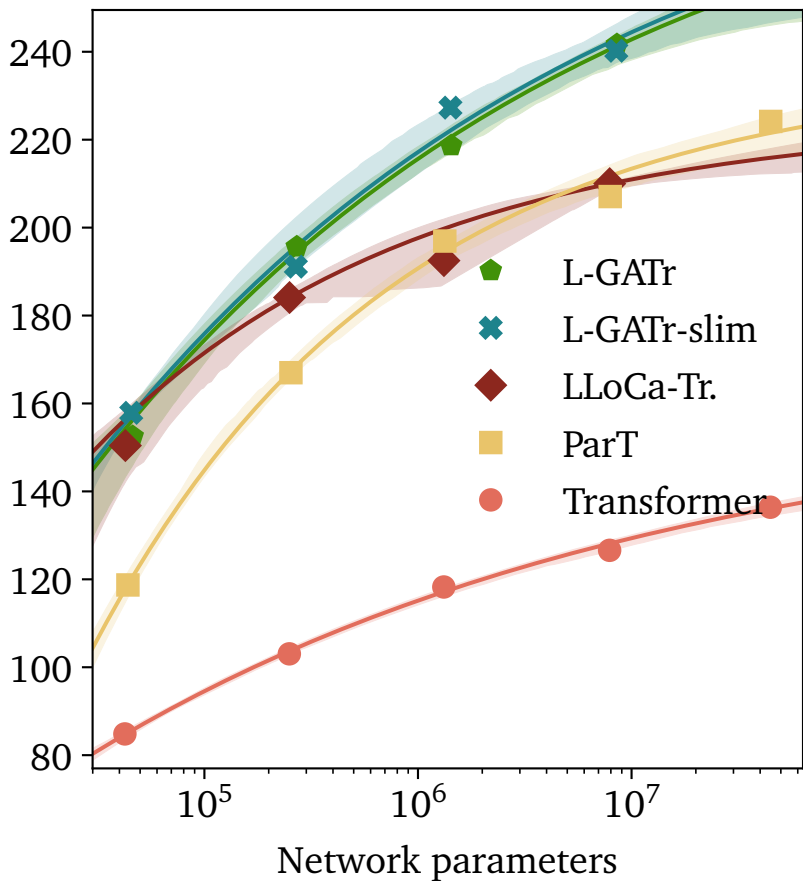
$\epsilon_{\text{bkg}}^{-1}$ @ $\epsilon_{\text{sig}} = 0.5$, cluster energy scale down

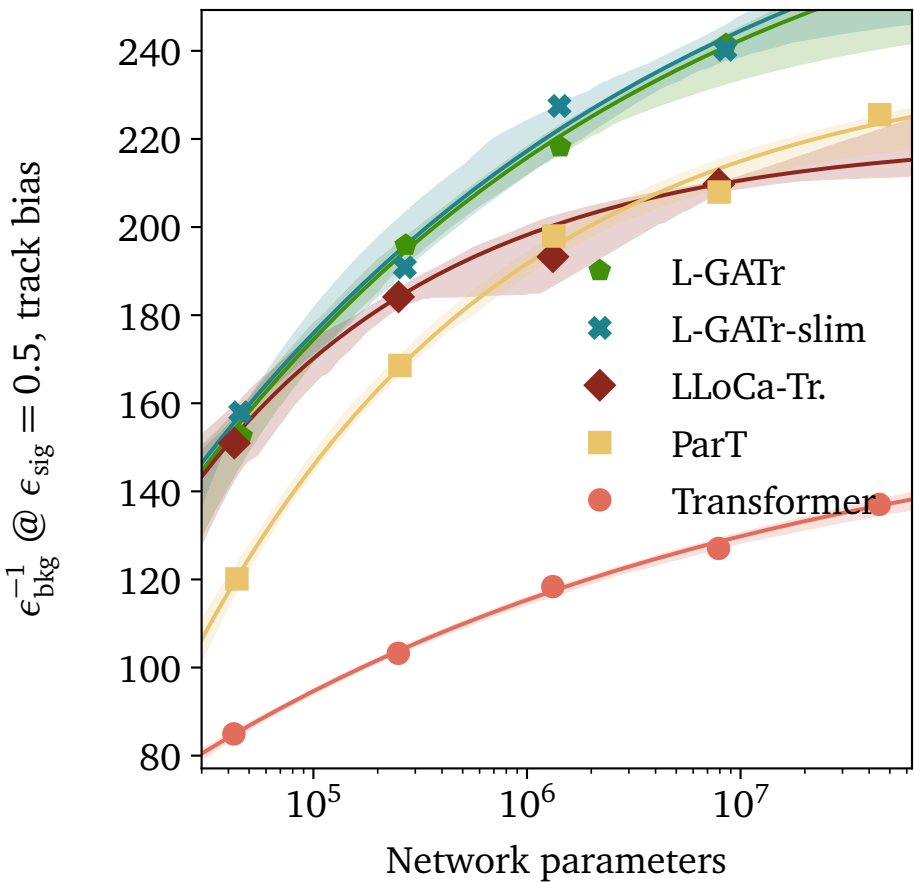


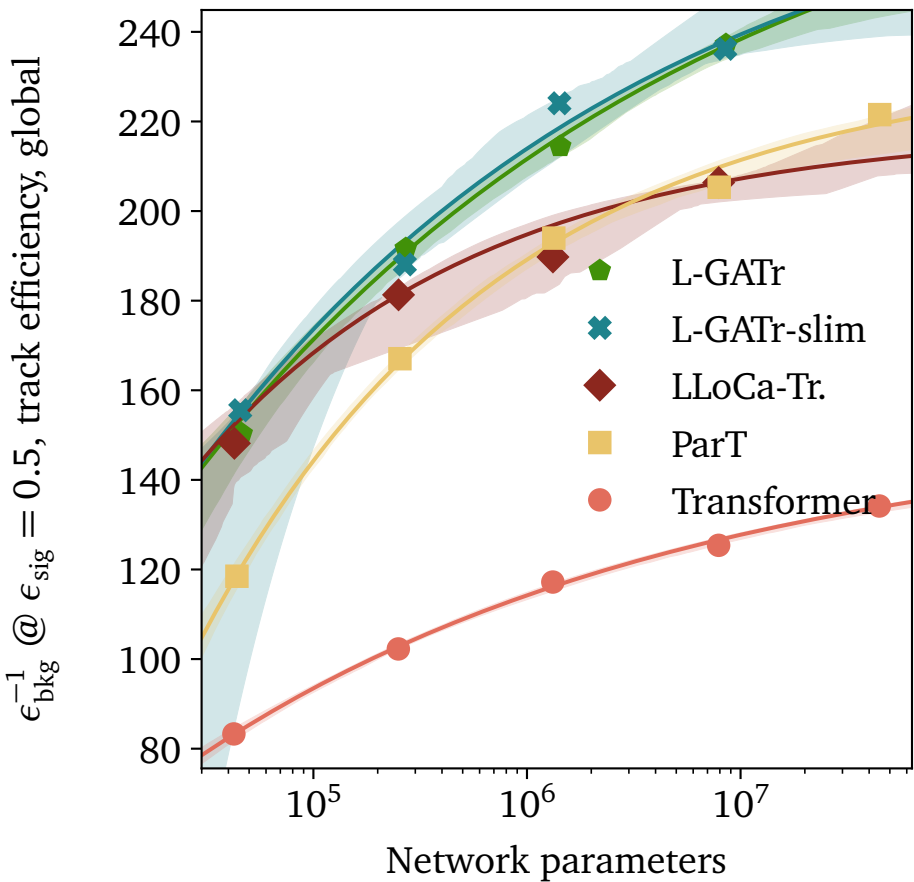
$\epsilon_{\text{bkg}}^{-1}$ @ $\epsilon_{\text{sig}} = 0.5$, cluster energy resolution

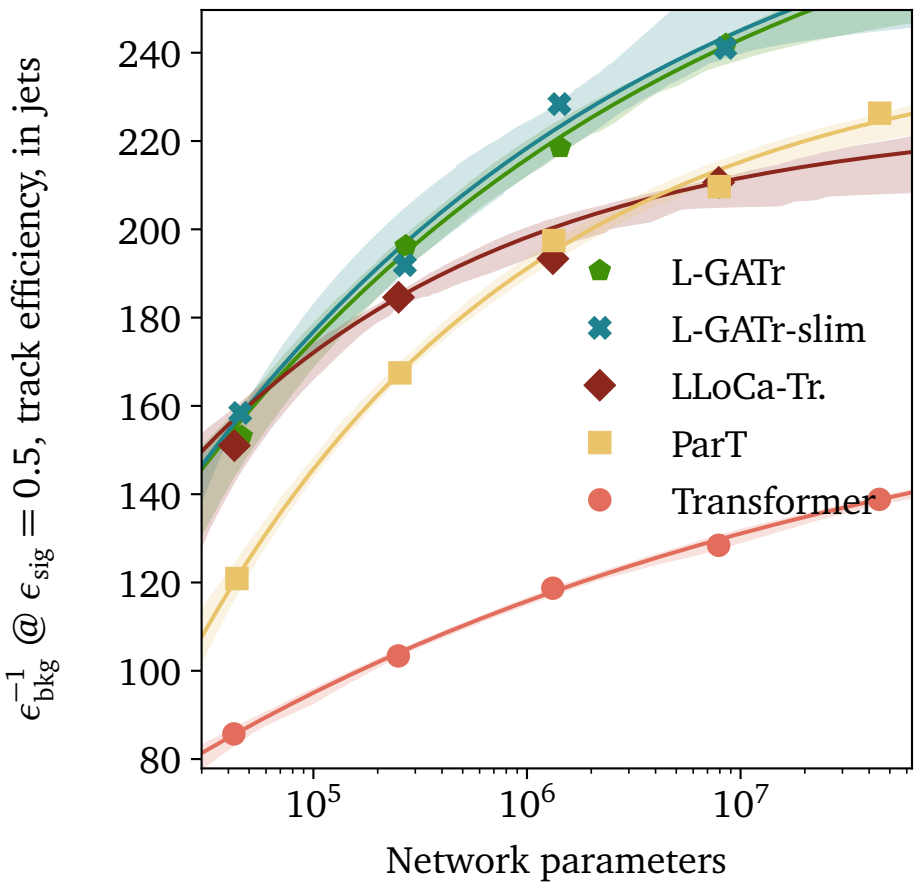


$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5$, cluster position

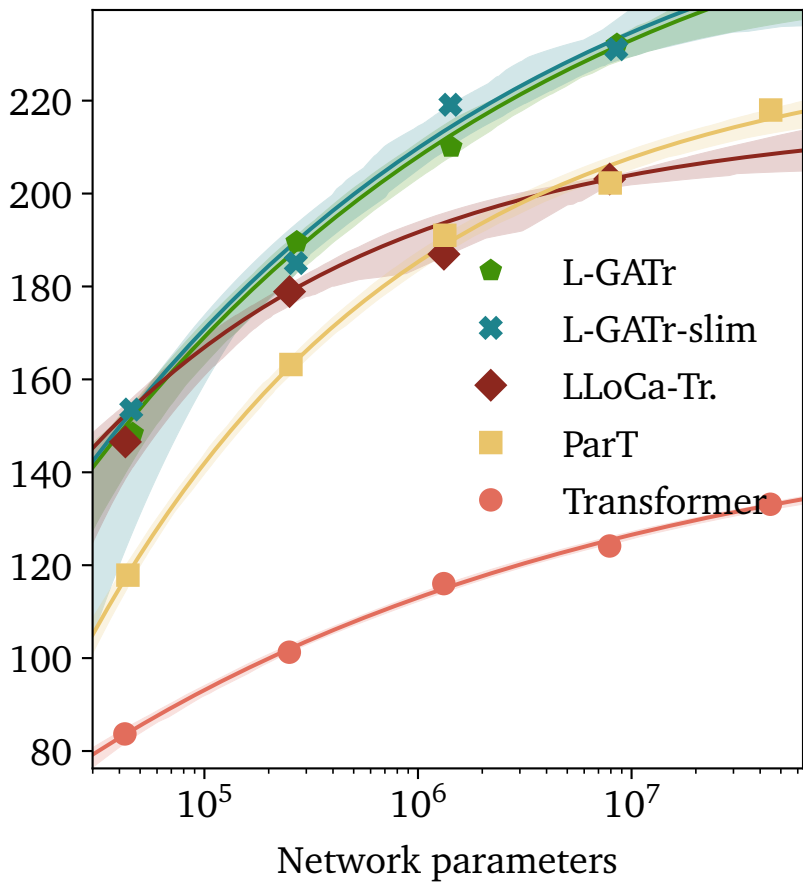




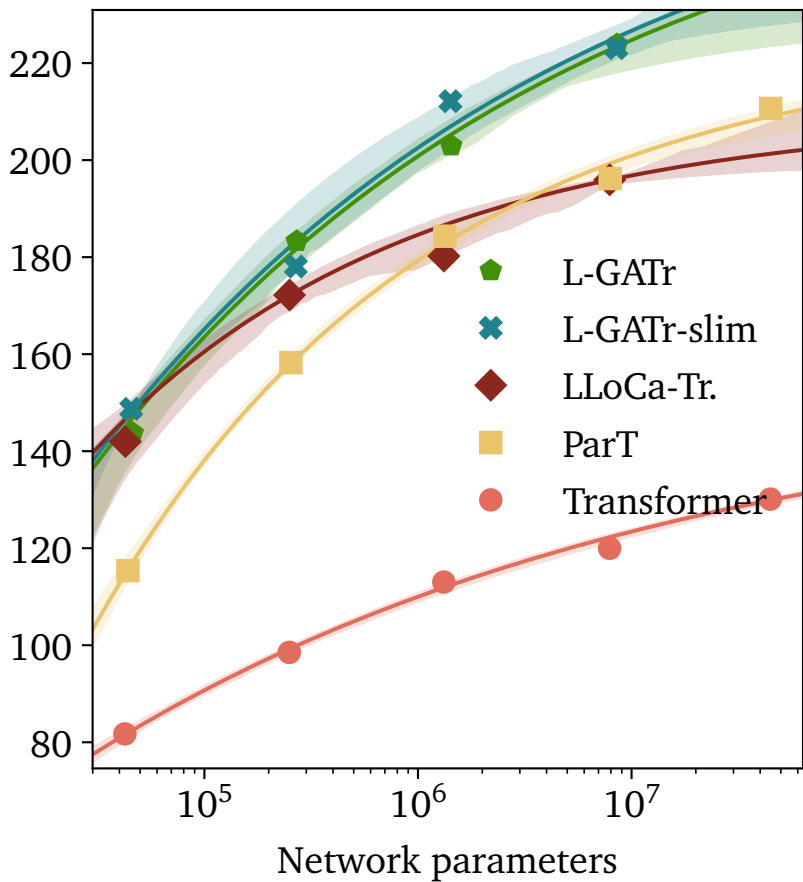




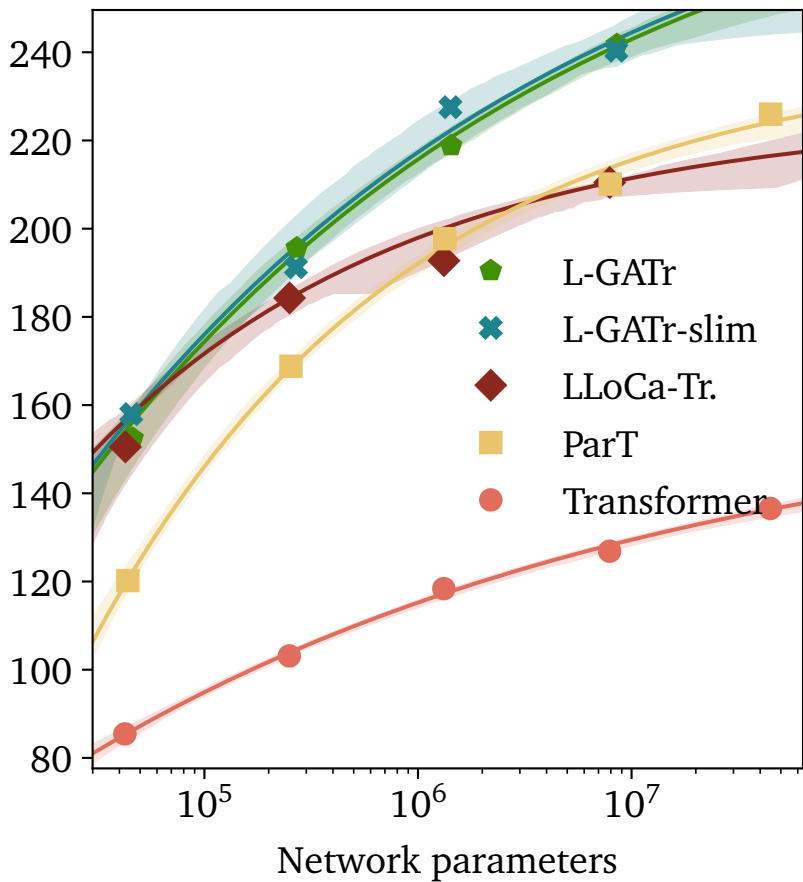
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ track fake rate, loose}$



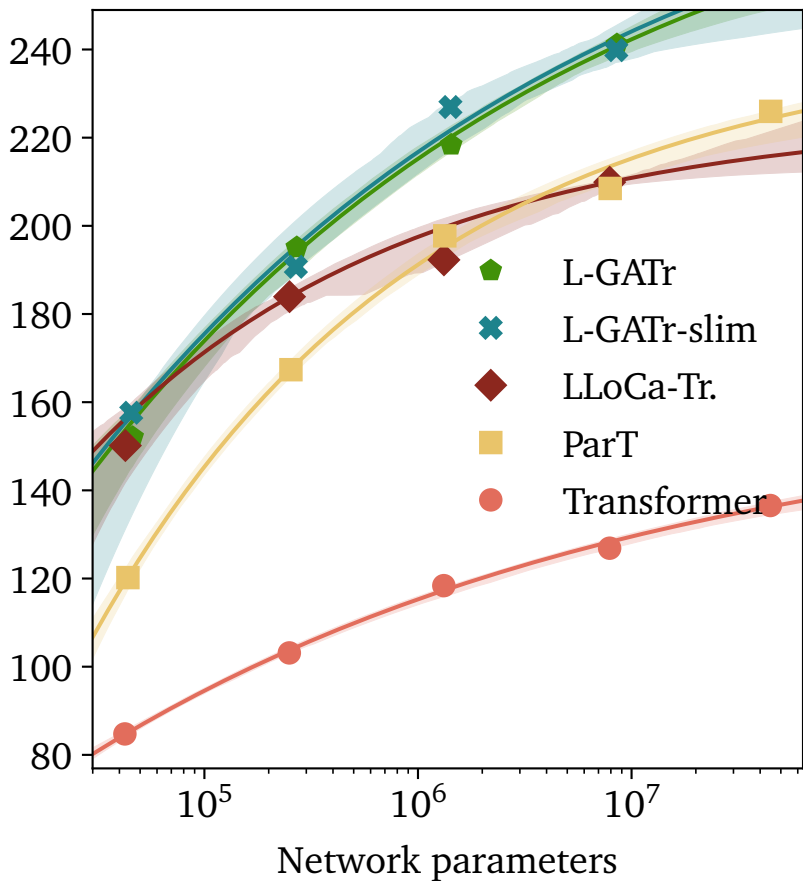
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5$, track fake rate, in jets



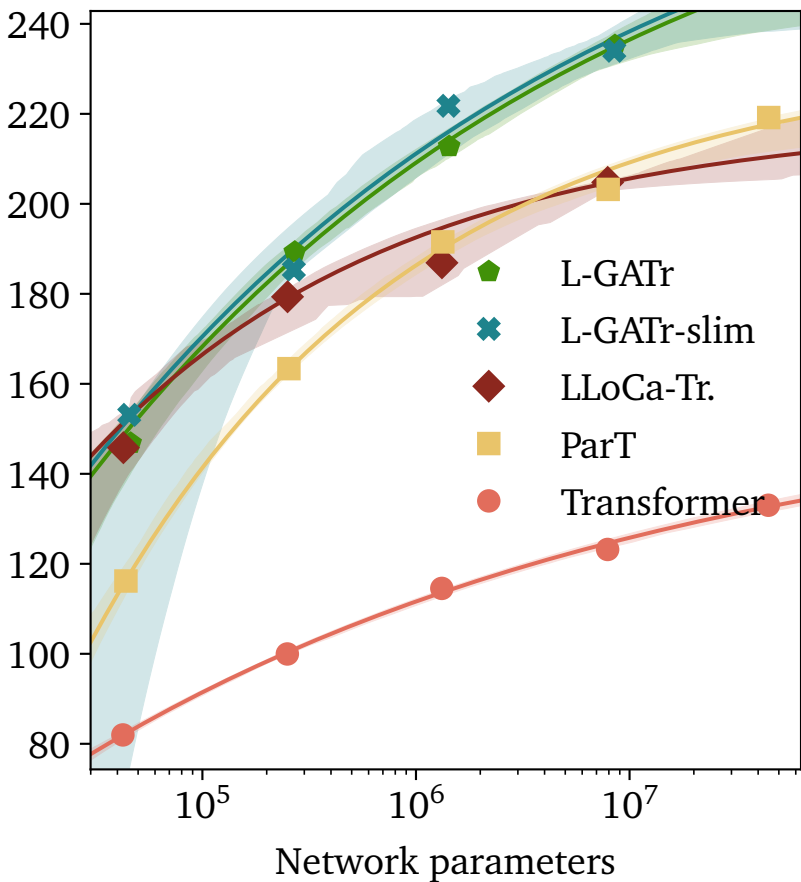
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ signal ISR} \times 2$



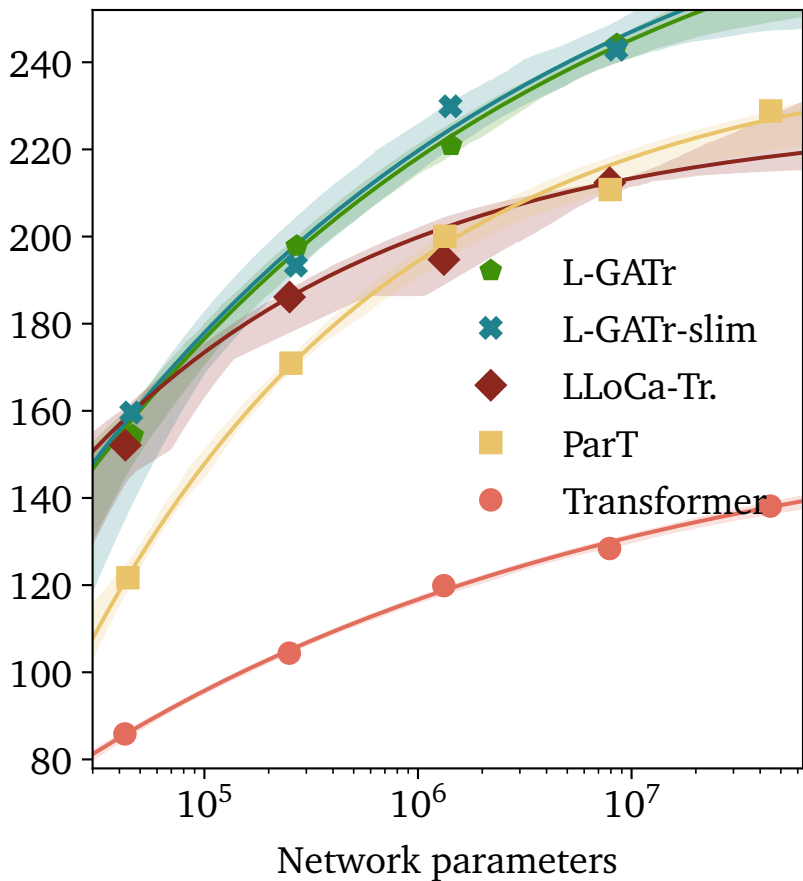
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ signal ISR} \times 0.5$



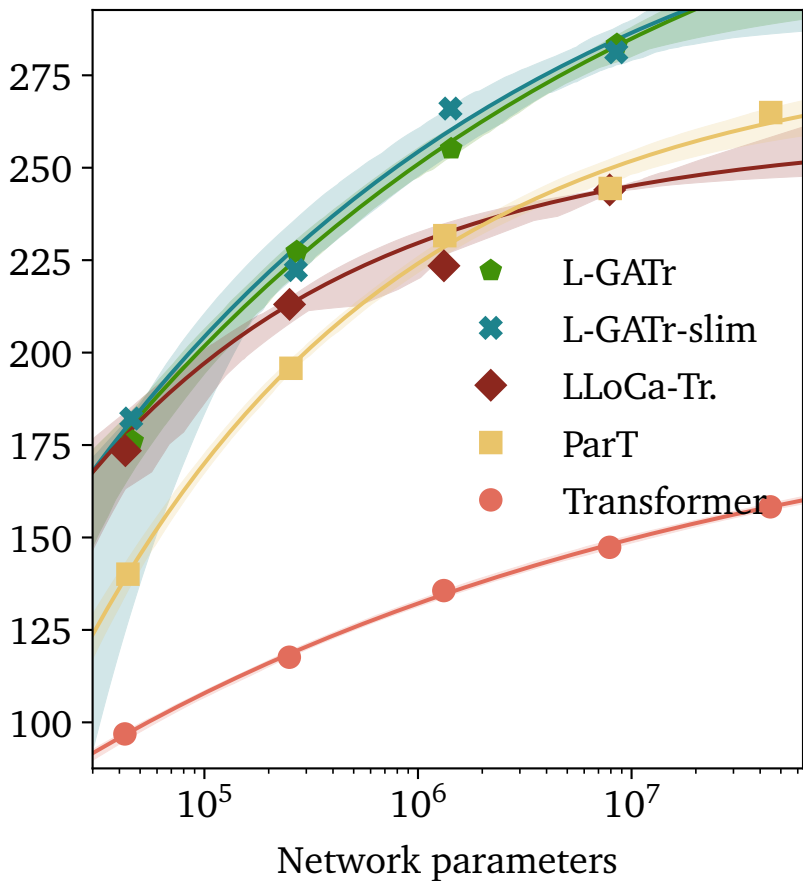
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ background ISR} \times 2$



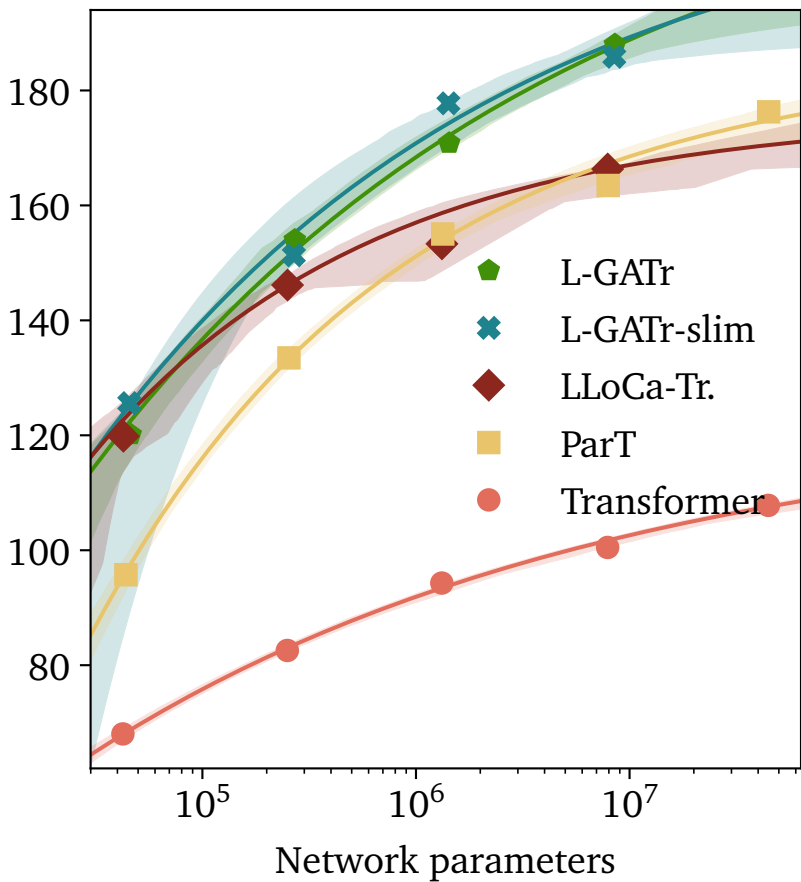
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ background ISR} \times 0.5$



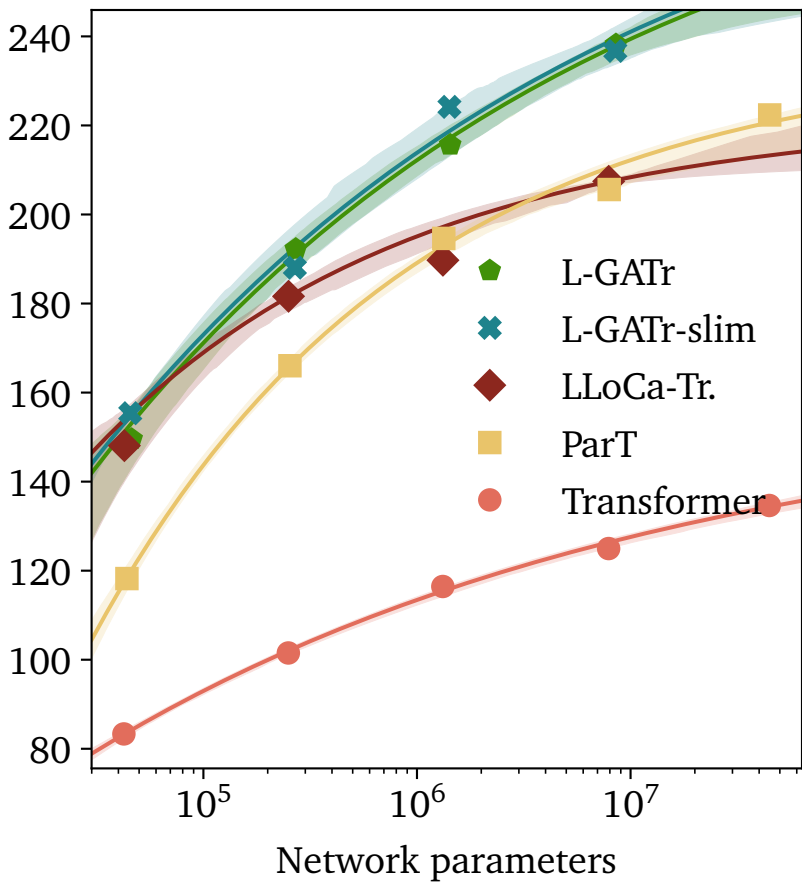
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ signal FSR} \times 2$



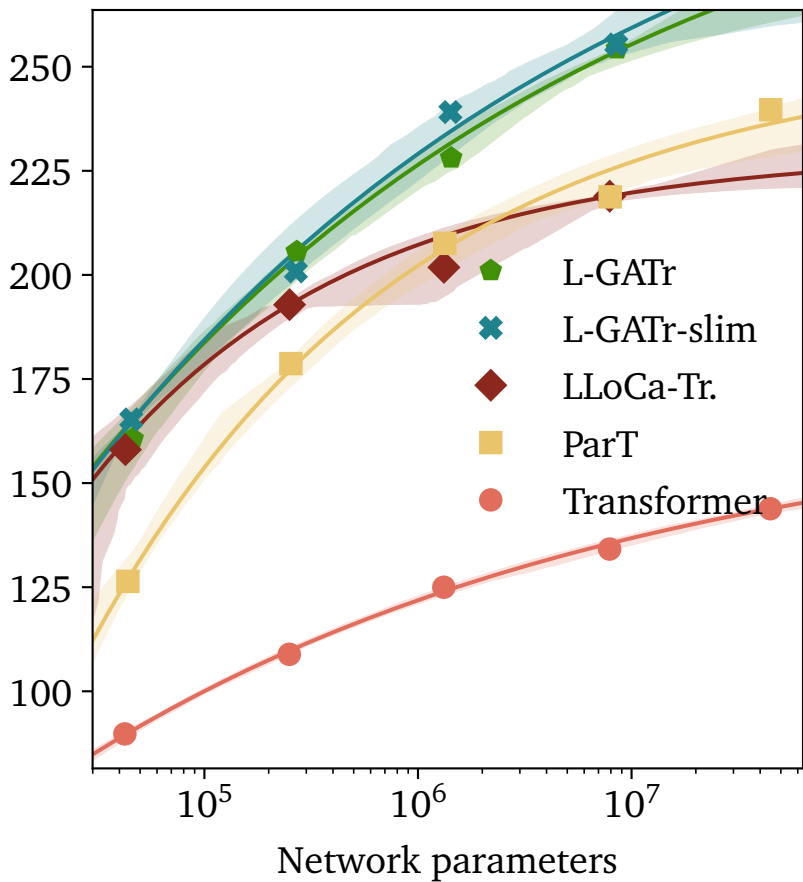
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ signal FSR} \times 0.5$

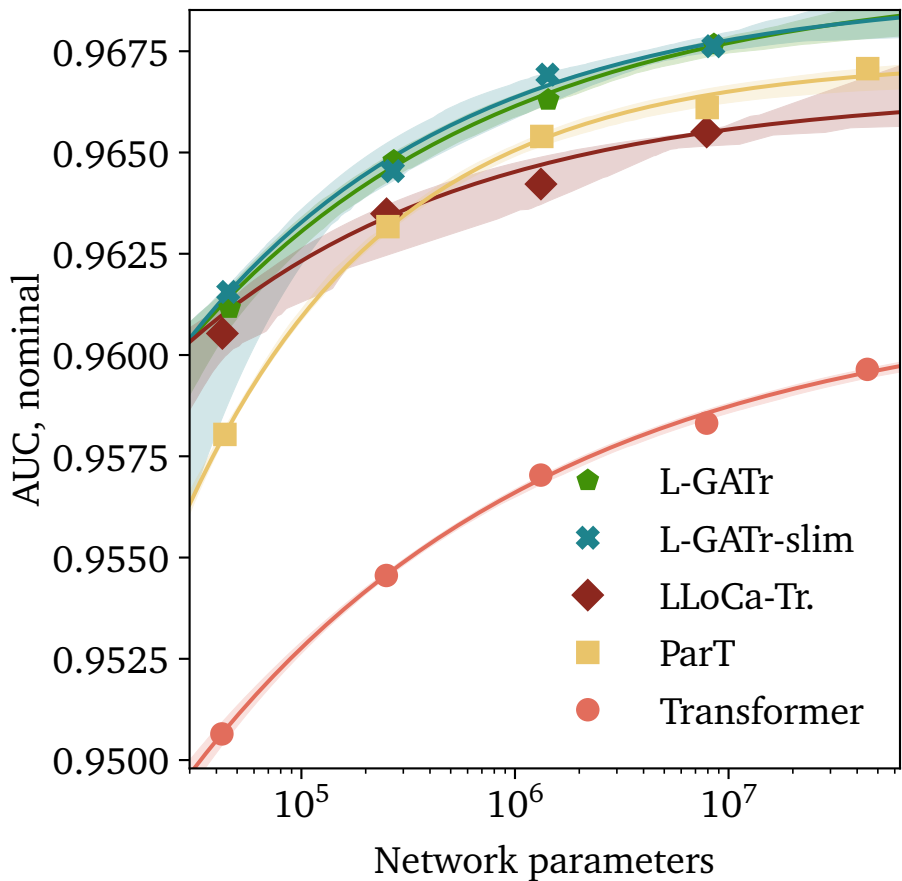


$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ background FSR} \times 2$

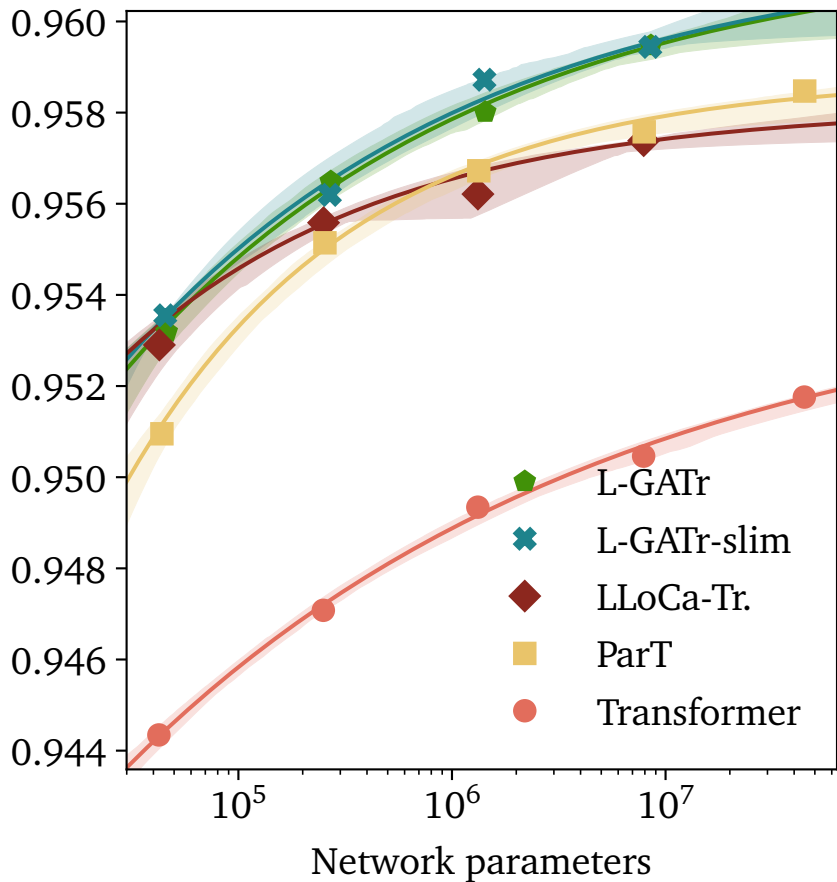


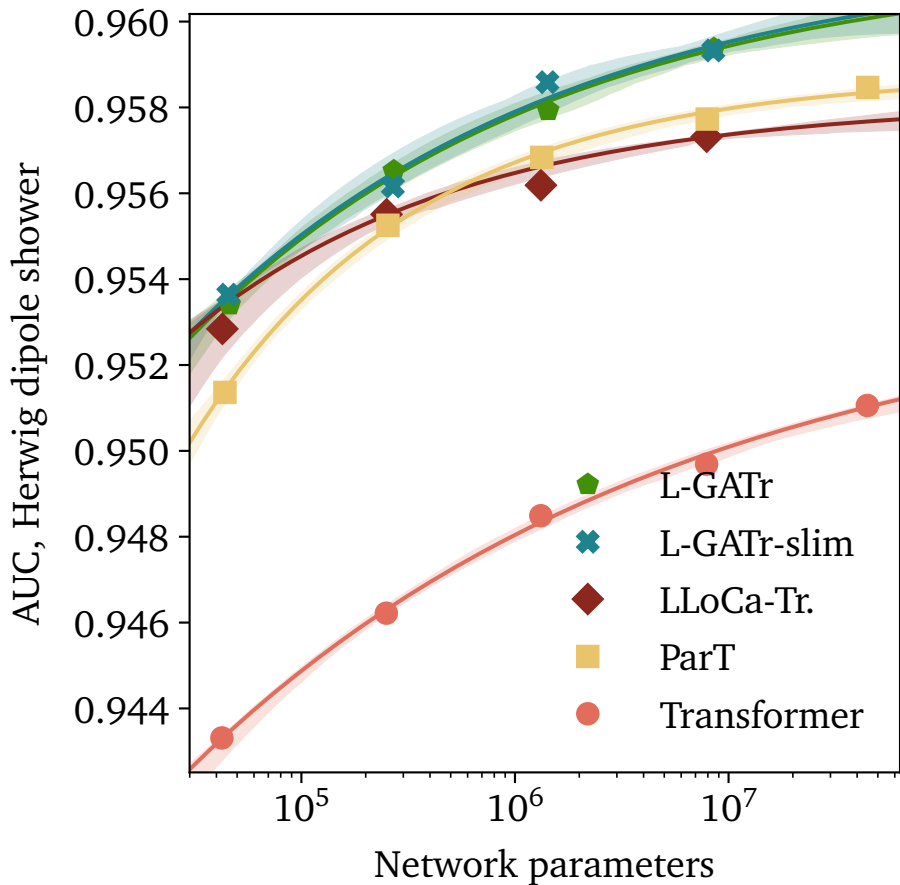
$\epsilon_{\text{bkg}}^{-1} @ \epsilon_{\text{sig}} = 0.5, \text{ background FSR} \times 0.5$

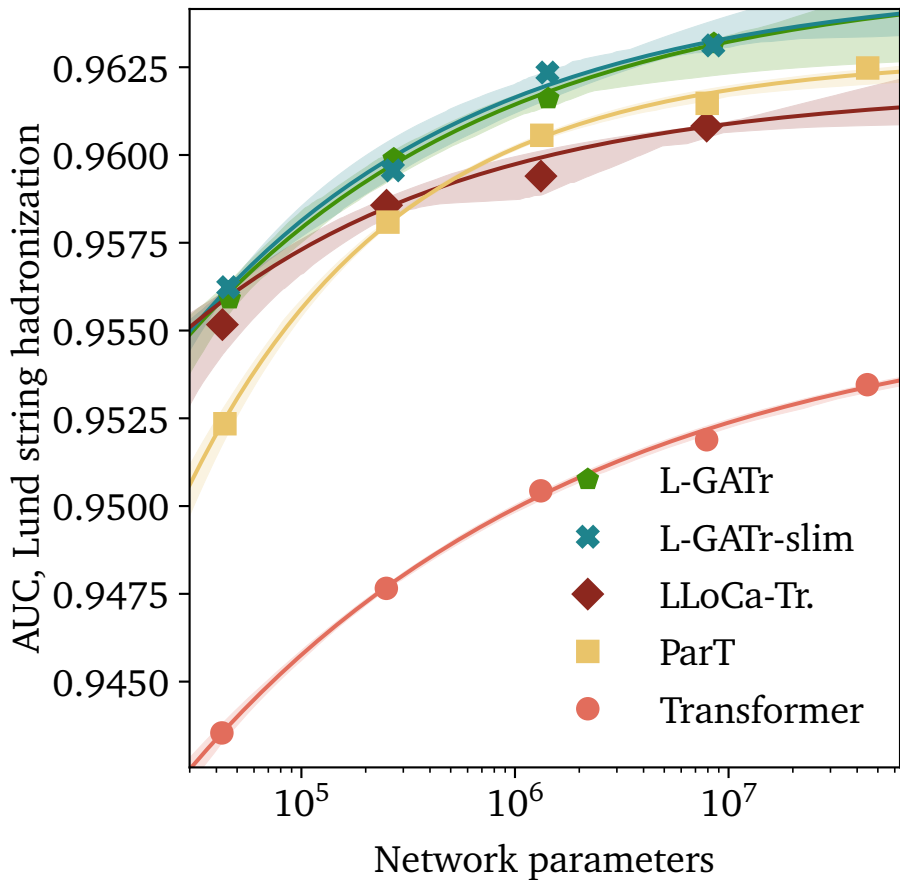


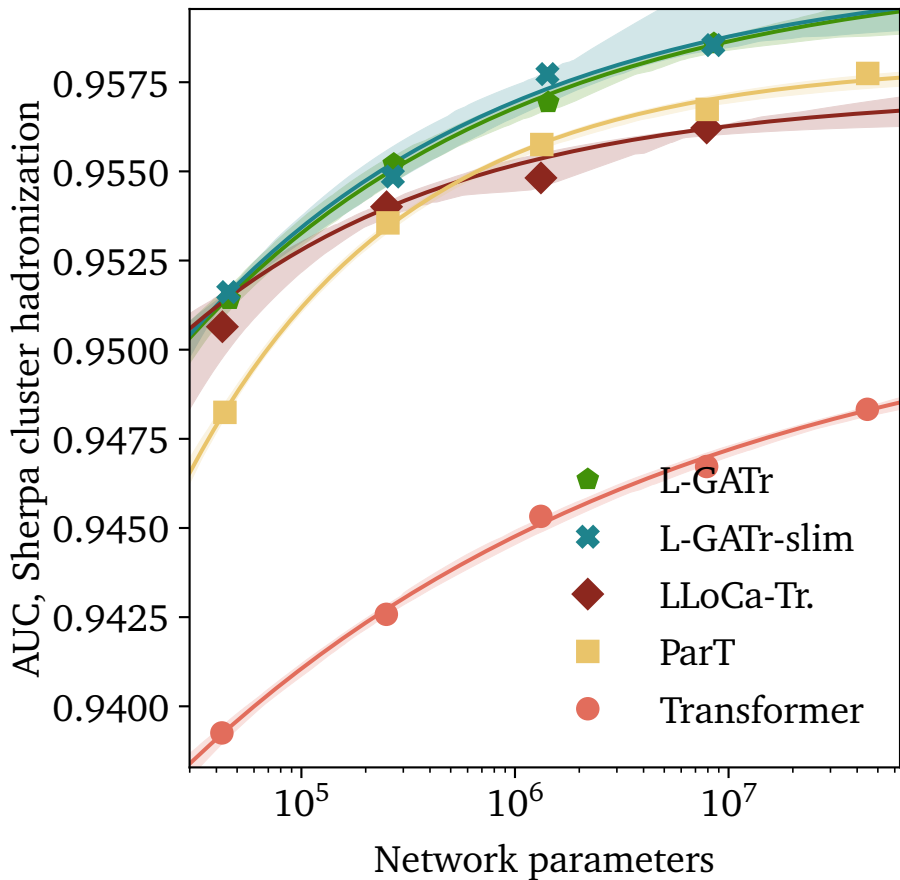


AUC, Herwig angular-ordered shower









AUC, Pythia $t\bar{t}$ signal

